

DECOYSHIELD

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CERTIFICATE

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DECLARATION

We hereby declare that the project report **DECOYSHIELD**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Ms.Anu Treesa George**.

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Abstract

DecoyShield is a cybersecurity system designed to improve network security using honeypot-based deception techniques. The system deploys decoy ports that imitate real network services to attract attackers and divert them away from actual systems. By interacting with these decoy services, attackers reveal their behavior and attack methods in a controlled environment. The system continuously monitors incoming network connections to detect suspicious activities such as port scanning, brute-force attacks, and unauthorized access attempts.

When attackers interact with the system, important information such as IP addresses, accessed ports, and timestamps is recorded. This data is stored in a database for further analysis and helps administrators understand attack patterns. The system also provides a monitoring dashboard that displays alerts, logs, and attack statistics in real time.

DecoyShield includes adaptive deception capabilities that allow it to deploy additional decoy services when suspicious activity increases. The system is implemented using Python, PyQt5, and SQLite, making it lightweight and easy to deploy. Overall, DecoyShield provides a practical and cost-effective solution for monitoring cyber threats and protecting real network resources..

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Chapter 1

Introduction

1.1 Overview

DecoyShield is a cybersecurity system designed to enhance network security using deception techniques. It is based on the concept of honeypots that attract attackers to fake services. These fake services imitate real network services. The main goal is to protect real systems from cyber attacks. Cyber threats such as hacking and unauthorized access are increasing rapidly. Traditional security systems sometimes detect attacks only after damage occurs. Because of this, proactive security solutions are needed. DecoyShield provides such a proactive approach to cybersecurity. The system deploys decoy ports that look like real network services. Attackers scanning the network may interact with these fake ports. This interaction helps the system capture attacker behavior safely. The real system remains hidden from attackers. The system continuously monitors incoming network traffic. It identifies suspicious activities such as port scanning. It can also detect brute-force login attempts. When suspicious behavior is detected, the system records the details. Information such as IP addresses and ports is stored. The system also records timestamps of the attack attempts. All these details are stored in a database for analysis. This data helps administrators understand attack patterns. The system uses rule-based analysis to study attacker behavior. This analysis helps determine the type of attack. The system can

classify the severity of threats. Based on attacker behavior, the system may deploy additional decoy services. This adaptive behavior increases the effectiveness of the honeypot. It keeps attackers engaged for a longer time. The longer engagement allows the system to collect more data. The collected data can be used for cybersecurity research. DecoyShield also includes an administrator dashboard. The dashboard provides real-time monitoring of network activity. Administrators can view alerts and attack reports through it. The dashboard displays attack logs and system status. It helps administrators analyze trends in cyber attacks. The interface is simple and easy to use. This makes it suitable for both researchers and security administrators. The system is developed using modern programming tools. Python is used for backend development and network monitoring. Streamlit is used to build the dashboard interface. SQLite is used as the database for storing logs. These technologies are lightweight and efficient. The system can run on a normal computer or server. It does not require expensive hardware. This makes it cost-effective and easy to deploy. DecoyShield is especially useful in academic environments. Students can use it to study cyber attack behavior. Organizations can use it to improve their security awareness. The system protects real resources by diverting attackers. It also collects valuable cybersecurity intelligence. Overall, DecoyShield is a practical and effective security solution. It demonstrates how deception technology can strengthen modern network security.

1.2 General Background

Cybersecurity has become a critical aspect of modern information systems. With the rapid growth of internet technologies, computer networks are widely used in many fields. Organizations depend on digital systems to store and manage important data. As a result, protecting these systems from cyber attacks has become very important. Cyber

attacks can cause data loss, financial damage, and disruption of services. Attackers use different techniques to exploit vulnerabilities in networks. Some common attack methods include hacking, malware, and unauthorized access. Port scanning is often used by attackers to identify open services in a network. Brute-force attacks are used to guess passwords repeatedly. These techniques help attackers gain access to sensitive systems.

Traditional security tools such as firewalls and intrusion detection systems are used to protect networks. Firewalls control incoming and outgoing network traffic. Intrusion detection systems monitor network activities for suspicious behavior. However, these tools sometimes fail to detect new or unknown attacks. Many security systems depend on known attack signatures. Attackers can bypass these systems using new attack methods. Therefore, researchers are developing new security approaches. One such approach is the use of honeypots in cybersecurity.

A honeypot is a decoy system designed to attract attackers. It mimics real services and appears vulnerable to intruders. When attackers interact with the honeypot, their activities are monitored. This helps security experts study attacker behavior safely. Honeypots also help in detecting new threats and attack patterns. They provide valuable information for improving security strategies. As cyber threats continue to evolve, deception-based security systems are becoming increasingly important..

1.3 Problem Defenition

Modern computer networks face increasing cyber threats due to the rapid growth of internet technologies. Attackers often attempt unauthorized access through techniques such as port scanning and brute-force attacks. Traditional security systems like firewalls and intrusion detection systems are commonly used to protect networks.

However, these systems often detect attacks only after suspicious activities have already occurred. This delay in detection can lead to data breaches and system damage. Therefore, there is a need for more proactive methods to improve network security.

Another major challenge in cybersecurity is the emergence of new and unknown attack techniques. Attackers continuously develop advanced methods to bypass traditional security systems. Signature-based detection systems are often unable to detect these new threats. Attackers also perform reconnaissance activities to identify vulnerabilities in a network. If these activities are not detected early, they may lead to serious cyber attacks. Hence, systems that can monitor and analyze attacker behavior are necessary.

To overcome these challenges, a system is required that can both protect networks and study attacker activities. The system should divert attackers away from real services to a controlled environment. It should monitor suspicious connections and record important attack information. The collected data can help administrators understand attack patterns. This information can also be used to improve future security strategies. Therefore, a deception-based cybersecurity solution can play an important role in strengthening network protection.

1.4 Scope of the system

The scope of the DecoyShield system is to enhance network security by using cyber deception techniques. The system focuses on protecting real network services by diverting attackers toward decoy services. These decoy services imitate real network ports and applications to appear legitimate to attackers.

- **Cyber Deception Security:** Enhancing network security by diverting attackers from real network services to decoy services that imitate legitimate ports and

applications.

- Suspicious Activity Detection: Monitoring incoming network traffic to identify activities such as port scanning, repeated login attempts, and unauthorized connection requests.
- Attack Data Logging: Recording attacker information such as IP addresses, targeted ports, timestamps, and action sequences for security analysis.
- Secure Data Storage: Storing captured attack data safely in a database so administrators can study attack patterns and detect vulnerabilities.
- Real-Time Monitoring: Providing a dashboard that displays alerts, logs, and attack details, helping administrators track suspicious activities and respond quickly.
- Behavior Analysis and Research: Allowing analysis of attacker interactions with decoy services and supporting cybersecurity research in a controlled environment.
- Lightweight and Compatible System: Designing the system to be easy to deploy on standard computers and work alongside security tools like firewalls and intrusion detection systems.

1.5 Objective

The primary objective of the DecoyShield system is to enhance network security through the use of cyber deception techniques. The detailed objectives are as follows:

- The system aims to protect real network services from attackers by diverting them to decoy services that imitate real network ports and applications.

- These decoy services make attackers believe they are interacting with legitimate systems, thereby reducing the risk of attackers reaching critical network resources.
- Another important objective is to continuously monitor incoming network traffic for suspicious activities such as port scanning, brute-force login attempts, and repeated unauthorized access requests.
- When such activities occur, the system captures detailed information about the attacker.
- The system records important details including IP addresses, targeted ports, timestamps, and interaction patterns to help administrators analyze attacker techniques.
- Another objective is to securely store all captured attack data for future analysis and investigation.
- The system also provides real-time monitoring through an administrator dashboard that displays alerts, logs, and statistics related to attack activities, allowing administrators to respond quickly.
- DecoyShield aims to increase attacker engagement using adaptive deception strategies, enabling attackers to interact with the decoy system for a longer time and provide more useful information.
- The system supports cybersecurity research and education, improves threat detection methods, and provides a practical, lightweight, and cost-effective solution for improving network security and cybersecurity awareness.

Chapter 2

Literature Review

2.1 Multi-Tier Honeypot for Resilient Network Security – Salama et al [1]

Salama et al. proposed a multi-tier honeypot architecture to strengthen network security against advanced cyber attacks. The system integrates three levels of honeypots: low-interaction, medium-interaction, and high-interaction honeypots. Each level provides different degrees of interaction with attackers, allowing the system to capture detailed information about malicious activities. The architecture is further enhanced by incorporating dual firewalls and the Snort Intrusion Prevention System (IPS). Suspicious network traffic is automatically redirected to the honeypot environment for analysis. This approach helps in identifying different types of attacks such as Distributed Denial of Service (DDoS) and reconnaissance attacks. The layered honeypot structure improves the ability to monitor and analyze attacker behavior at multiple levels. It also helps security administrators understand how attackers interact with network services. However, the deployment of multiple honeypots and security layers increases system complexity. The architecture also requires higher system resources and careful configuration for effective operation.

2.2 Simulation and Performance Testing of the Ganesha Honeypot System (GHOST) for SSH Security – Saskara et al. [2]

Saskara et al. introduced the Ganesha Honeypot System (GHOST), which focuses on monitoring attacks targeting Secure Shell (SSH) services. The system is built as a medium-interaction honeypot based on the Cowrie honeypot platform. It diverts attackers to a simulated SSH environment that behaves like a real server. When attackers attempt to log in, the system allows them to access a virtual shell. Their activities, including login attempts and executed commands, are recorded for analysis. The system captures detailed information about brute-force attacks commonly used against SSH servers. To evaluate the performance of the system, the researchers conducted experiments using the Hydra password-cracking tool. The results showed that GHOST effectively captured attack behavior while maintaining stable system performance. The system also provided real-time alerts and logging capabilities. However, the main limitation of the system is that it focuses only on SSH attacks and does not cover other network services.

2.3 Honeypot Allocation for Cyber Deception Under Uncertainty – Ahmed H. Anwar et al. [3]

Ahmed H. Anwar et al. proposed a strategic honeypot allocation model for improving cyber deception in uncertain environments. The study focuses on dynamically placing honeypots within a network to confuse and mislead attackers. The authors use a two-phase allocation strategy based on game theory and reinforcement learning

techniques. In this approach, cyber deception is modeled as a Partially Observable Markov Decision Process (POMDP). This model helps determine optimal honeypot placement even when the defender has limited information about attacker behavior. The system adapts its deception strategy based on observed attacker actions. By dynamically adjusting honeypot placement, the system increases the probability of detecting malicious activities. The model also improves attacker confusion and delays their progress within the network. This approach enhances the overall effectiveness of cyber defense strategies. However, the computational complexity of the model makes real-time implementation challenging.

2.4 HoneyDecoy: A Comprehensive Web-Based Parasitic Honeypot System for Enhanced Cybersecurity

– Hexuan Fan et al. [4]

Hexuan Fan et al. proposed HoneyDecoy, a web-based parasitic honeypot system designed to improve cybersecurity monitoring. The system integrates honeypot functionality directly with existing web servers. Instead of creating separate fake systems, the honeypot mirrors real web services to make the deception more realistic. HoneyDecoy uses behavior analysis to understand how attackers interact with web applications. The system also applies a Markov Decision Process (MDP) to dynamically generate honeypots based on attacker actions. This dynamic generation improves the realism and effectiveness of the deception. Attackers interacting with the system believe they are accessing legitimate services. The system records detailed logs of web-based attacks and suspicious activities. This information helps administrators trace attacker behavior and identify vulnerabilities in web applications. However,

the system mainly focuses on HTTP and web-based attacks and requires careful synchronization with production servers.

2.5 Adaptive Honeypots: Dynamic Deception Tactics for Cyber Defense [5]

This research presents an adaptive honeypot framework designed to improve cyber defense using dynamic deception strategies. The system continuously adjusts honeypot behavior based on attacker interactions. Reinforcement Learning (RL) techniques are used to analyze attacker behavior and update the honeypot configuration. This allows the honeypot to become more realistic and difficult for attackers to detect. Adaptive honeypots can dynamically change services, responses, and interaction levels. This increases the amount of time attackers spend interacting with the honeypot system. Longer engagement allows security teams to gather more detailed intelligence about attack techniques. The framework also helps detect previously unknown attack patterns. By adapting to evolving cyber threats, the system improves overall network defense capability. However, designing effective rules and training reinforcement learning models can be complex and resource-intensive.

No.	Description	Advantages	Disadvantages
1	A multi-tier honeypot framework combining low, mid and high-interaction honeypots with dual firewalls and Snort IPS to improve proactive and postintrusion network defense.	- Layered honeypot structure improves detection of complex attacks like DDoS and reconnaissance. - Integration with firewalls and Snort IPS strengthens proactive and post-intrusion defense.	- Deployment and configuration are complex due to multiple security layers. - Requires higher system resources and ongoing maintenance
2	An active, medium-interaction SSH honeypot based on Cowrie that captures brute-force attacks and logs attacker activity with real-time alerts .	- Effectively captures detailed SSH brute-force behavior including login attempts and commands. - Provides real-time alerts and visual monitoring, improving incident response speed.	- Focuses only on SSH services and does not cover other network attacks. - Medium-interaction design limits deeper attacker engagement.
3	This paper presents a dynamic honeypot allocation strategy that adapts deception techniques under uncertain attacker behavior to improve defensive effectiveness.	- Adaptive and strategic honeypot placement - Improves attacker confusion and deception.	- Computationally intensive. - Difficult to apply in real-time environments.
4	A web-based parasitic honeypot that integrates with production servers to mirror real web services and capture detailed information about web-based attacks.	- Realistic deception for web attackers. - Minimal impact on legitimate services.	- Limited to HTTP/web attacks. - Requires careful synchronization.
5	A rule-based adaptive honeypot system that dynamically changes its behavior and exposed services based on attacker interaction to increase engagement and deception.	- No machine learning dependency. - Improves attacker engagement time.	- Rule design is complex. - Limited scalability for advanced attacks.

Table 2.1: Comparison of Existing Literature and DecoyShield

Chapter 3

Requirement Specification

Requirement specification describes the hardware, software, and functional components required to develop and operate the **DecoyShield system**. It defines the resources and technologies needed to implement the honeypot-based cybersecurity solution. The specification ensures that the system can effectively monitor network traffic, deploy decoy services and capture attacker activities. It also helps developers understand the technical environment in which the system will operate. By clearly defining these requirements, the system can be developed in an efficient and structured manner. The requirement specification also ensures that the system remains lightweight, reliable and suitable for academic and small organizational environments.

3.1 Hardware Requirements

Hardware requirements refer to the physical components needed to run the DecoyShield system.

- **Computer or Server System:** A computer or server is required to host and run the DecoyShield application. It executes the monitoring program, manages decoy services and processes incoming network traffic. The system acts as the central environment where attacker activities are analyzed.

- **Stable Network Connection:** A reliable network connection is necessary for monitoring incoming network traffic. The system continuously observes network connections to detect suspicious activities such as port scanning or repeated access attempts.
- **Storage Capacity:** Adequate storage is required to store attack logs and system data. The system records information such as IP addresses, timestamps and accessed ports. These logs are used for analyzing attacker behavior and improving security strategies.

3.2 Software Requirements

Software requirements refer to the programs and tools needed to implement and run the system.

- **Operating System:** The system can run on operating systems such as Windows or Linux that support Python and networking tools.
- **Honeypot Service:** The honeypot service creates decoy ports and fake services that appear real to attackers. These services attract attackers and allow the system to safely monitor their activities
- **Network Monitoring Tools:** Network monitoring tools observe incoming traffic and identify suspicious behavior. They help detect activities such as port scanning and unauthorized access attempts.
- **Database System:** A database system is required to store attack logs and system information. SQLite is used because it is lightweight and easy to integrate with Python applications.

- **Admin Monitoring Dashboard:**The dashboard provides a graphical interface where administrators can monitor attacks, view logs and analyze security alerts in real time.

3.3 Programming Languages and Tools

Software requirements refer to the programs and tools needed to implement and run the system.

- **Python:**Python is used as the main programming language for developing the system. It handles network monitoring, decoy service creation and attacker behavior analysis.
- **Streamlit:**Streamlit is used to develop the dashboard interface. It allows administrators to view attack logs, alerts and system status through an interactive interface.
- **SQLite:**SQLite is used as the database to store attack logs and system data. It provides efficient data storage without requiring a separate database server.

3.4 Functional Requirements

No.	Requirement	Description
FR1	Deployment of Decoy Ports	The system must create fake network ports that appear as real services to attackers.
FR2	Monitoring Network Traffic	The system continuously monitors incoming network connections to detect suspicious activities.
FR3	Zero-Day Threat Block	The system must automatically block any process that deviates from established safe behavioral baselines, such as unauthorized encryption attempts.
FR4	Attacker Behavior Analysis	The system analyzes attacker interactions with decoy services to identify different types of cyber attacks.
FR5	Logging Attack Data	All attacker activities are recorded and stored in the database for future analysis.
FR6	Alerts and Reports	The system provides alerts and reports through the dashboard to help administrators monitor security threats.

Table 3.1: DECOYSHIELD Functional Requirements

3.5 Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance standards that the DecoyShield system must satisfy. These requirements focus on how the system operates rather than the functions it performs. They ensure that the system is secure, reliable, efficient and easy to use.

No.	Requirement	Description
NFR1	Performance	The system should monitor network traffic continuously and detect suspicious activities without delay. It must handle multiple connection attempts and record attacker interactions efficiently.
NFR2	Security	The system must securely store attack logs and protect them from unauthorized access. Only authorized administrators should be able to access the dashboard and system data.
NFR3	Reliability	The system should operate continuously and maintain accurate logging of attacker activities. It must provide stable performance while monitoring network traffic.
NFR4	Usability	The system should provide a simple and user-friendly dashboard for administrators. The interface should clearly display alerts, logs, and system status.
NFR5	Scalability	The system should support increasing network traffic and additional decoy services without affecting performance.
NFR6	Maintainability	The system should be easy to maintain and update. The design should allow developers to modify or improve the system when needed.

Table 3.2: DECOYSHIELD Non-Functional Requirements

Chapter 4

Proposed System and Design

4.1 Proposed System

The proposed system, **DecoyShield**, is an adaptive cybersecurity solution that uses honeypot-based deception techniques to improve network security. Instead of exposing real services directly to the internet, the system deploys decoy ports and fake services that imitate real network resources. These decoy services are designed to attract attackers and divert them away from actual systems. When attackers attempt to access the network, they interact with these decoy services instead of real services.

The system continuously monitors incoming network traffic to detect suspicious activities such as port scanning, repeated access attempts and brute-force login attacks. When an attacker interacts with the decoy environment, the system records important information including IP address, accessed ports, timestamps and command activities. This information is stored securely for further analysis.

DecoyShield also analyzes attacker behavior using rule-based techniques to identify patterns of malicious activity. If suspicious activity increases, the system can dynamically deploy additional honeypots to improve deception and capture more information about the attack. The collected data is displayed through an administrator dashboard that allows security administrators to monitor attack activities in real time. By combining deception, monitoring, and behavior analysis, the proposed

system helps protect real network resources while providing valuable cybersecurity intelligence.

4.1.1 Key Components

The DecoyShield system consists of several key components that work together to detect and analyze cyber attacks.

1. **Connection Monitoring Module:** This module continuously monitors incoming network traffic and detects connection attempts from external sources. It identifies suspicious patterns such as port scanning or repeated login attempts. The module acts as the first layer of detection in the system.
2. **Decoy Service Module:** The decoy service module creates fake ports and services that imitate real network applications. These services attract attackers and encourage them to interact with the system. By doing so, attackers are diverted away from actual network resources.
3. **Behavior Analysis Module:** This module analyzes attacker interactions with the decoy services. It studies patterns such as repeated connection attempts, login failures and suspicious commands. Based on this analysis, the system determines whether the activity is malicious.
4. **Logging and Database Module:** All attacker activities are recorded in a database. The system stores details such as attacker IP address, targeted ports, timestamps and interaction patterns. This stored information helps administrators analyze attack trends and improve security strategies.
5. **Admin Dashboard Module:** The admin dashboard provides a graphical interface for monitoring system activity. It displays alerts, attack logs and system

status in real time. Administrators can use this dashboard to track suspicious activities and analyze attacker behavior.

4.1.2 Overall Workflow

The operational flow of the NIVA system is designed to be transparent to the user while maintaining high security:

1. **Deployment of Decoy Services:** The system deploys fake ports and services that imitate real network services. These decoy services attract attackers and divert them away from actual systems.
2. **Continuous Network Monitoring:** The system continuously monitors incoming network traffic and detects connection attempts. It identifies suspicious activities such as port scanning and repeated access attempts.
3. **Attacker Interaction:** When an attacker connects to the network, the system redirects them to the decoy services. The attacker interacts with the fake environment believing it to be real.
4. **Capturing Attack Data:** The system records important information such as attacker IP address, targeted ports, timestamps and login attempts during the interaction.
5. **Behavior Analysis:** The captured data is analyzed to identify patterns of malicious activity and determine the type of attack.
6. **Logging and Storage:** All attack details are stored in a database for further analysis and security improvements.
7. **Admin Monitoring:** Administrators can view alerts, logs and attack details through a monitoring dashboard in real time.

4.2 Feasibility Study

A feasibility study evaluates whether the proposed system can be successfully developed and implemented. It analyzes different aspects such as technical capability, operational practicality and economic viability. For the DecoyShield project, the feasibility study ensures that the system can be developed using available technologies, operated effectively by administrators and implemented with minimal cost.

4.2.1 Technical Feasibility

Technical feasibility examines whether the required technology, tools, and resources are available to develop the DecoyShield system. The system is developed using widely available technologies such as Python, PyQt5 and SQLite, which are easy to implement and widely supported. Python provides strong networking libraries that help in monitoring network traffic and deploying decoy ports. PyQt5 is used to create the monitoring dashboard that allows administrators to visualize attack data in real time. SQLite is used as a lightweight database to store attack logs and system information.

The system does not require specialized hardware and can run on a standard computer or server with a stable network connection. The development of the system requires basic knowledge of networking concepts and cybersecurity techniques. Since these technologies and skills are readily available in academic environments, the implementation of DecoyShield is technically feasible.

4.2.2 Operational Feasibility

Operational feasibility evaluates how effectively the system can function in a real environment and whether it will be useful for its intended users. The DecoyShield

system is designed to be simple and easy to operate. Administrators can monitor network activity through a user-friendly dashboard that displays alerts, attack logs and system status. This interface allows users to observe suspicious activities and analyze attack patterns easily.

The system works alongside traditional security tools such as firewalls and intrusion detection systems. It does not interfere with normal network operations because attackers interact only with decoy services instead of real systems. The system also supports manual monitoring and decision-making by administrators. Because of its simplicity and practical design, the DecoyShield system is operationally feasible for both academic and small organizational environments.

4.2.3 Economic Feasibility

Economic feasibility evaluates whether the cost of developing and maintaining the system is reasonable. The DecoyShield system is designed to be cost-effective because it uses open-source technologies. Tools such as Python, PyQt5 and SQLite are freely available and do not require licensing fees. The system also does not require expensive hardware infrastructure since it can run on standard computers.

The development cost is minimal because the project mainly involves software implementation. Maintenance costs are also low because the system is lightweight and easy to update. Compared to commercial cybersecurity solutions that require expensive licenses and hardware, DecoyShield provides an affordable alternative for monitoring cyber threats. Therefore, the system is economically feasible for academic institutions and small organizations.

4.3 Design

System design describes the overall structure of the DecoyShield system and how different components work together to detect and analyze cyber attacks. The design focuses on protecting real network services by diverting attackers to decoy services. It ensures efficient monitoring of network traffic, analysis of attacker behavior, and secure storage of attack data.

4.3.1 Module-Level Design

Connection Monitoring Module

The connection monitoring module continuously monitors incoming network traffic and connection attempts. It listens for requests coming from external sources and identifies suspicious patterns such as port scanning or repeated login attempts. This module acts as the first level of detection in the system and ensures that all incoming connections are analyzed

Decoy Service Module

The decoy service module creates fake network ports and services that imitate real network applications. These services appear legitimate to attackers and encourage them to interact with the system. By interacting with these decoy services, attackers are diverted away from real network resources.

Behavior Analysis Module

The behavior analysis module examines the actions performed by attackers while interacting with the decoy environment. It analyzes patterns such as repeated connection attempts, unusual commands or abnormal login behavior. This helps the

system identify different types of cyber attacks and determine the level of threat.

4.3.2 Architecture Diagram

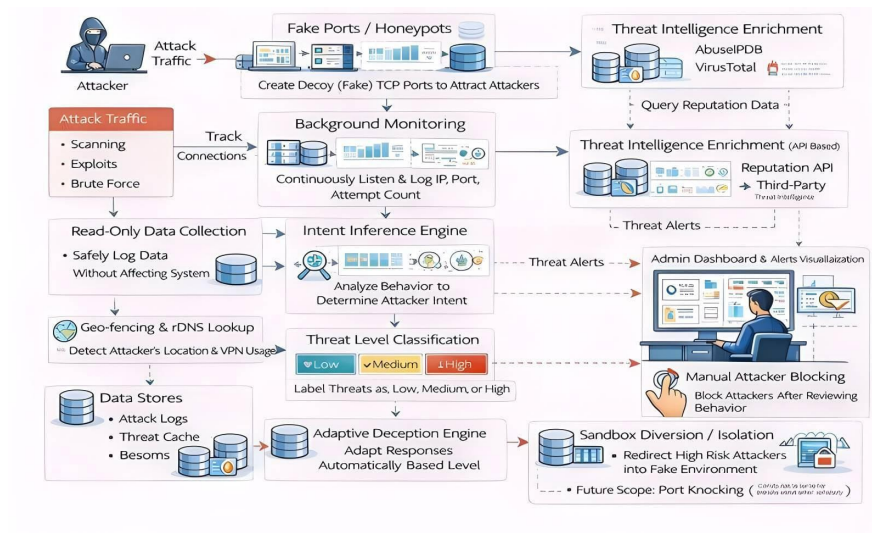


Figure 4.1: System Architecture

4.3.3 Use Case Diagram

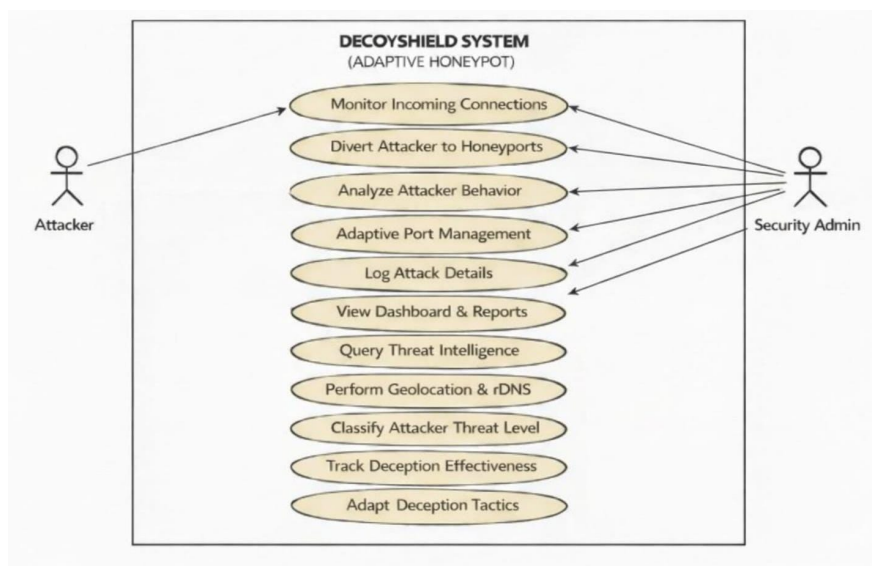


Figure 4.2: DecoyShield Use Case

4.3.4 Data Flow Diagram

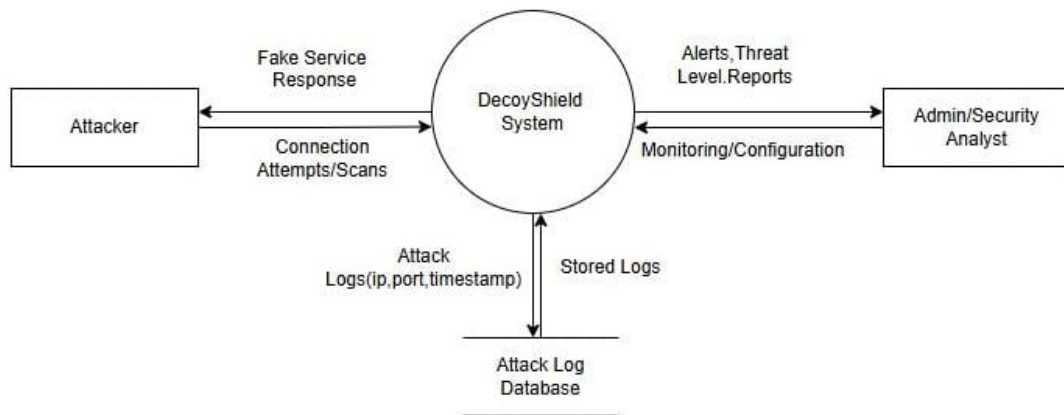


Figure 4.3: Data Flow-Level 0

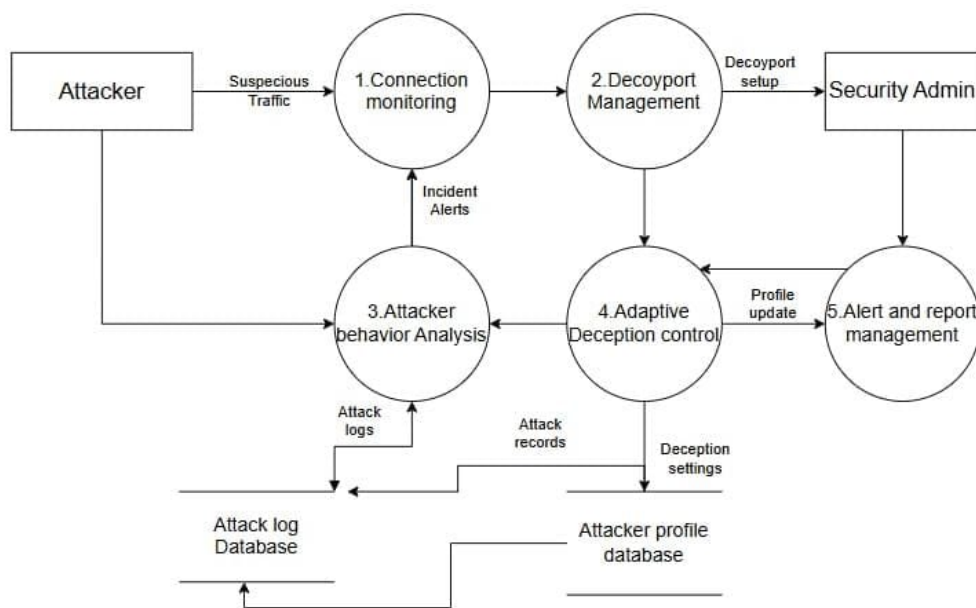


Figure 4.4: Data Flow-Level 1

4.3.5 Gantt Chart

- The figure shows the Gantt chart of the Decoy Shield project schedule.
- It displays the timeline of tasks from January 2026 to March 2026.
- Each bar represents the duration of a specific project activity.
- The chart shows important stages such as environment setup, module development, system integration, testing, and documentation.
- It also indicates the sequence and dependency between tasks.
- The project ends with final review and report submission.

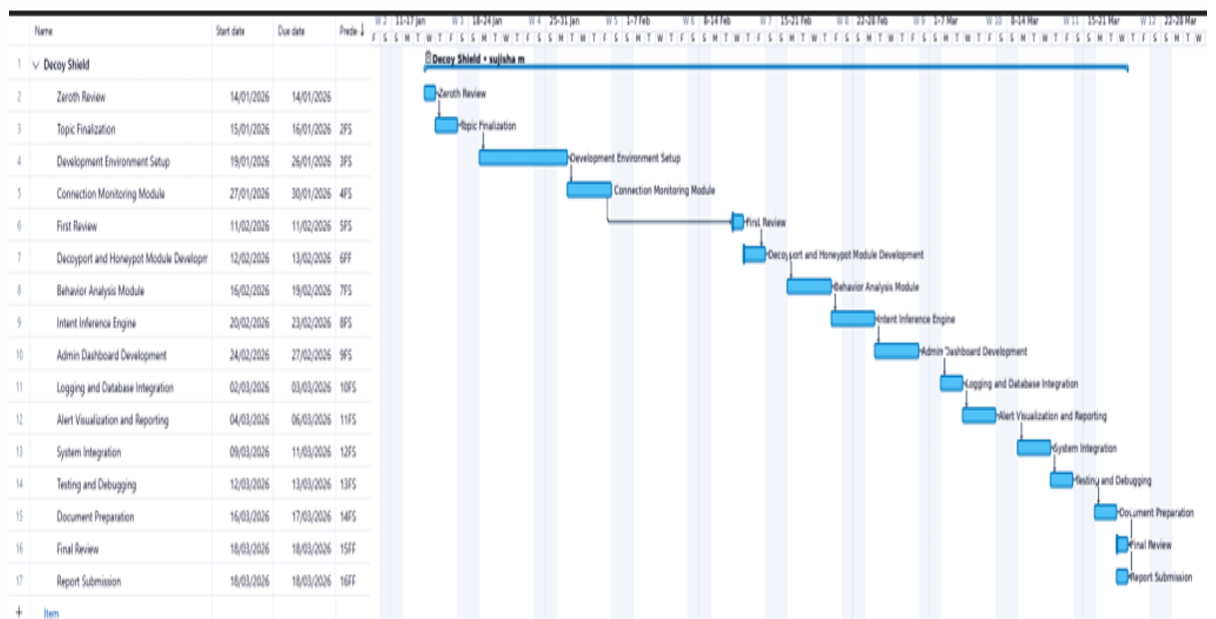


Figure 4.5: Gantt Chart

Chapter 5

Implementation

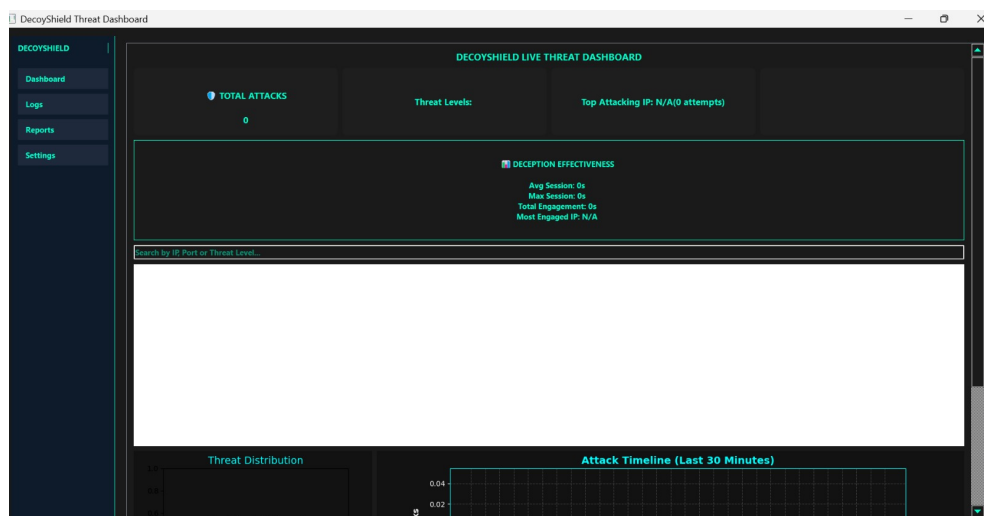


Figure 5.1: DecoyShield Threat Dashboard Interface



Figure 5.2: DecoyShield Live Threat Dashboard Overview

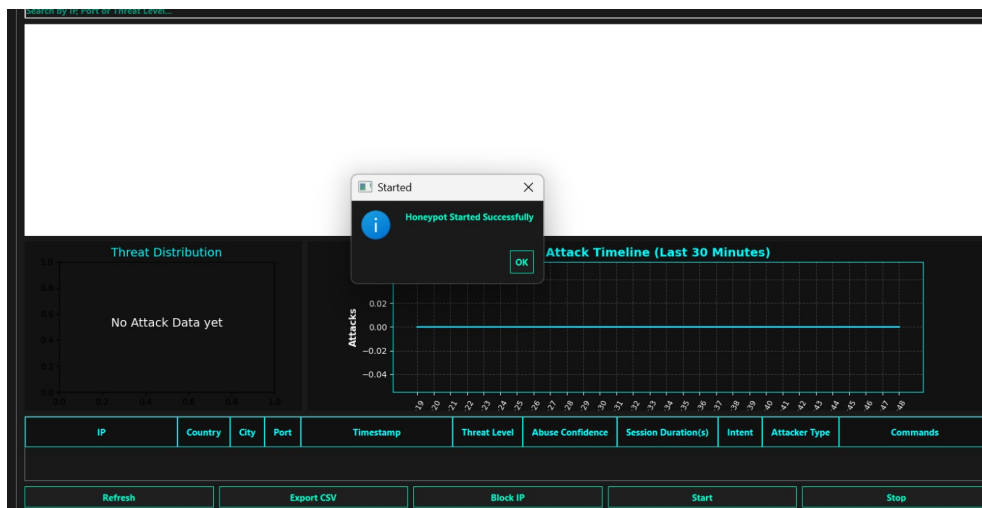


Figure 5.3: Honeypot Initialization Notification

```
PS C:\Users\nambi> nmap -sT -T5 -p- 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 20:59 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00086s latency).
Not shown: 65460 closed tcp ports (conn-refused)
PORT      STATE      SERVICE
21/tcp    open      ftp
22/tcp    open      ssh
23/tcp    open      telnet
25/tcp    open      smtp
53/tcp    open      domain
80/tcp    open      http
110/tcp   open      pop3
135/tcp   open      msrpc
137/tcp   filtered  netbios-ns
143/tcp   open      imap
443/tcp   open      https
445/tcp   open      microsoft-ds
2024/tcp  open      xinuexpansion4
2423/tcp  open      rnrp
3306/tcp  open      mysql
3336/tcp  open      directv-tick
3389/tcp  open      ms-wbt-server
3537/tcp  open      ni-visa-remote
3675/tcp  open      calltrax
4824/tcp  open      unknown
5040/tcp  open      unknown
5727/tcp  open      asgenf
5900/tcp  open      vnc
6123/tcp  open      backup-express
7654/tcp  open      unknown
7680/tcp  open      pando-pub
```

Figure 5.4: Detection of Initial Attack Activity

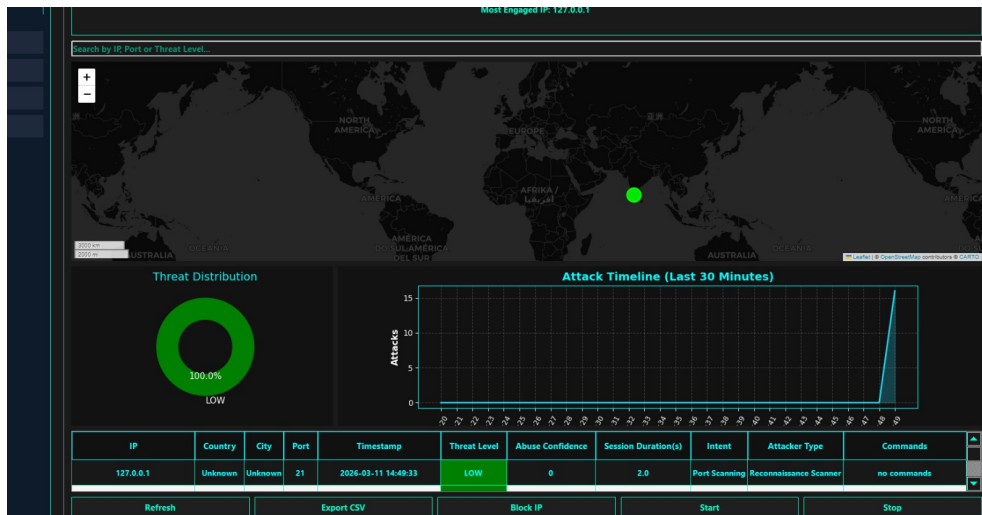


Figure 5.5: Critical Threat Alert Notification

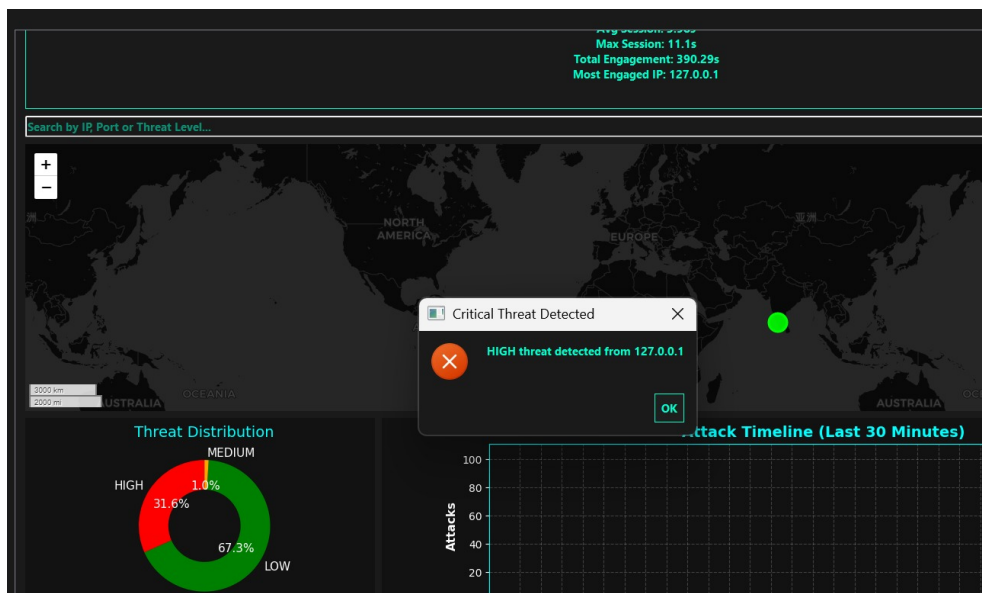


Figure 5.6: Threat Distribution Visualization

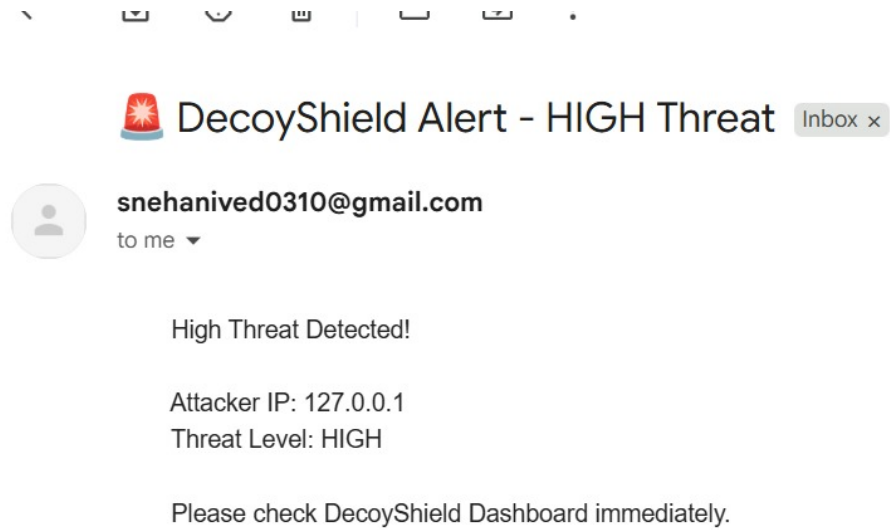


Figure 5.7: Email Alert for High-Severity Attack

ATTACK LOGS										
	IP	Country	City	Port	Time	Threat	Abuse Confidence	Session Duration(s)	Intent	Attacker Type
1	127.0.0.1	Unknown	Unknown	51204	2026-03-11 14:50:09	HIGH	0	5.85	HORIZONTAL PORT SCAN	Unknown Attacker
2	127.0.0.1	Unknown	Unknown	69	2026-03-11 14:50:07	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
3	127.0.0.1	Unknown	Unknown	10913	2026-03-11 14:50:07	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
4	127.0.0.1	Unknown	Unknown	10956	2026-03-11 14:50:07	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
5	127.0.0.1	Unknown	Unknown	1080	2026-03-11 14:50:07	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
6	127.0.0.1	Unknown	Unknown	43585	2026-03-11 14:50:07	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
7	127.0.0.1	Unknown	Unknown	40823	2026-03-11 14:50:07	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
8	127.0.0.1	Unknown	Unknown	22508	2026-03-11 14:50:07	HIGH	0	6.88	HORIZONTAL PORT SCAN	Unknown Attacker
9	127.0.0.1	Unknown	Unknown	32022	2026-03-11 14:50:07	HIGH	0	7.36	HORIZONTAL PORT SCAN	Unknown Attacker
10	127.0.0.1	Unknown	Unknown	43872	2026-03-11 14:50:07	HIGH	0	6.84	HORIZONTAL PORT SCAN	Unknown Attacker
11	127.0.0.1	Unknown	Unknown	22835	2026-03-11 14:50:06	HIGH	0	7.21	HORIZONTAL PORT SCAN	Unknown Attacker
12	127.0.0.1	Unknown	Unknown	31562	2026-03-11 14:50:05	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
13	127.0.0.1	Unknown	Unknown	19949	2026-03-11 14:50:05	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
14	127.0.0.1	Unknown	Unknown	161	2026-03-11 14:50:05	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
15	127.0.0.1	Unknown	Unknown	44470	2026-03-11 14:50:05	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
16	127.0.0.1	Unknown	Unknown	26111	2026-03-11 14:50:05	HIGH	0	7.49	HORIZONTAL PORT SCAN	Unknown Attacker
17	127.0.0.1	Unknown	Unknown	179	2026-03-11 14:50:04	LOW	0	2.0	Port Scanning	Reconnaissance Scanner
18	127.0.0.1	Unknown	Unknown	2082	2026-03-11 14:50:04	LOW	0	2.0	Port Scanning	Reconnaissance Scanner

Figure 5.8: Detailed Attack Logs Interface



Figure 5.9: Security Report Summary Dashboard

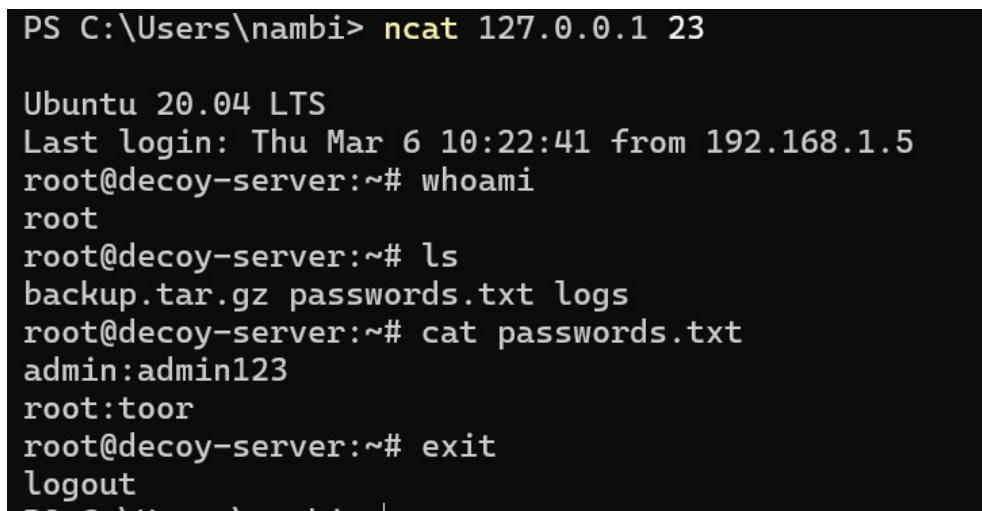


Figure 5.10: Simulated Attacker Interaction via Netcat

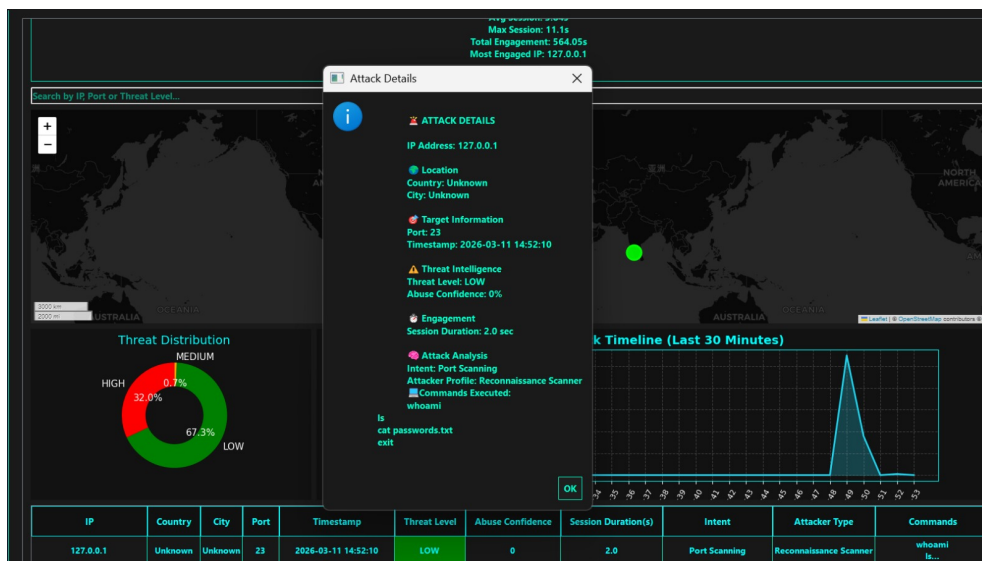


Figure 5.11: Attack Details Inspection Window

```

root@decoy-server: ~#
PS C:\Users\nambi> ncat 127.0.0.1 1111
PS C:\Users\nambi> ncat 127.0.0.1 2222
PS C:\Users\nambi> ncat 127.0.0.1 3333
PS C:\Users\nambi> ncat 127.0.0.1 80

[AUTHORIZED USER LOGIN ]
IP: 127.0.0.1
SERVICE: 80
TIMESTAMP: 2026-03-11 14:56:22
Welcome back, authorized user!

```

Figure 5.12: Multiple Service Probing Using Netcat

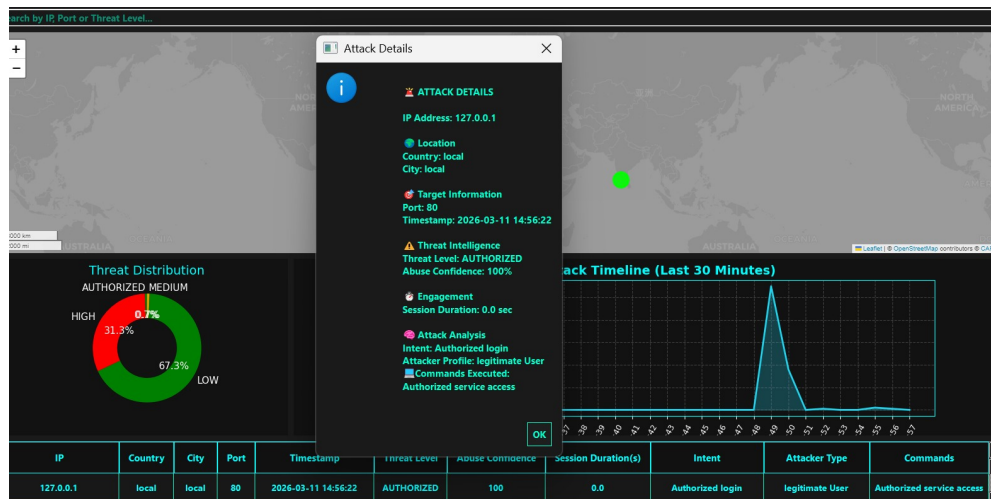


Figure 5.13: Authorized User Access Detection

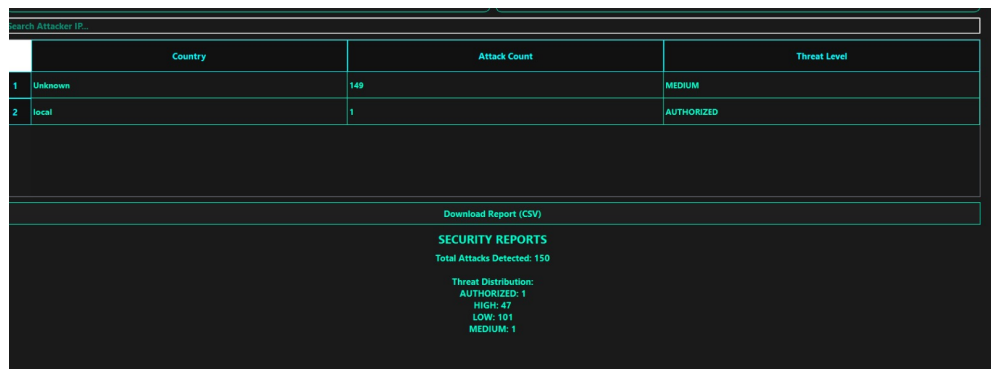


Figure 5.14: Final Security Report and Threat Statistics

Chapter 6

Results and Discussion

6.1 Result Overview

The DecoyShield system was successfully developed and implemented as an adaptive cyber deception solution designed to improve network security by diverting attackers toward decoy services instead of real network resources

6.2 Successful Deployment of Decoy Services

The system was able to deploy fake network ports (honeypots) that imitate real services. These decoy services successfully attract malicious users attempting activities such as port scanning, brute-force attacks and unauthorized access.

This allows the real production services to remain hidden and protected from direct attacks.

6.3 Real-Time Monitoring of Network Activity

DecoyShield continuously monitors incoming connections and records suspicious activities such as:

- Repeated connection attempts

- Port scanning behavior
- Unauthorized access attempts

The monitoring module captures and stores important attack details including IP address, targeted ports, timestamps and interaction patterns for further analysis.

6.4 Attacker Behavior Analysis

The system performs rule-based profiling to analyze attacker behavior. Based on the observed activity patterns, it can determine whether the traffic is malicious and classify the threat level.

This analysis helps administrators understand:

- Attacker strategies
- Frequency of attacks
- Commonly targeted services

6.5 Adaptive Honeypot Response

One of the key results of the project is the adaptive deception mechanism.

When suspicious behavior is detected, the system can:

- Dynamically open additional honeypots
- Increase attacker engagement
- Redirect malicious traffic away from real systems

This adaptive approach improves the effectiveness of cyber deception compared to static honeypots.

6.6 Attack Logging and Data Storage

All attack-related information is securely stored in a database for future analysis.

The system logs include:

- Attacker IP address
- Accessed ports
- Attack timestamps
- Attack type or threat level

These logs help in threat intelligence and forensic investigation.

6.7 Administrator Dashboard and Visualization

A lightweight monitoring dashboard was implemented to display:

- Attack statistics
- System status
- Logged attack records

This interface allows administrators to easily monitor ongoing threats and make informed security decisions.

6.8 Discussions

The DecoyShield system demonstrates how cyber deception can improve network security by diverting attackers toward decoy services instead of real systems. Instead of relying only on traditional security mechanisms, the system uses honeypots to attract and monitor malicious activities. The deployed decoy ports successfully imitate real network services and encourage attackers to interact with them. When attackers attempt actions such as port scanning or unauthorized access, the system records important details like IP address, access time and targeted ports.

These collected logs help administrators understand common attack patterns and behaviors. The adaptive deception mechanism is another important feature of the system. When suspicious activities such as repeated connection attempts are detected, the system dynamically activates additional honeypots. This increases attacker engagement and makes it harder for attackers to identify real network resources. The system also provides a monitoring dashboard that allows administrators to view alerts, attack logs and system status in real time.

This visualization improves the ability of administrators to respond quickly to potential threats. However, the current system mainly relies on rule-based analysis, which may not detect all advanced attack techniques. The system is also designed primarily for academic or small-scale environments. Large-scale deployment may require additional scalability and optimization. Despite these limitations, the system effectively combines deception, monitoring, and analysis to enhance network protection. Overall, DecoyShield provides a practical and cost-effective approach for understanding and managing cyber threats.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

The DecoyShield system was successfully developed as an adaptive cyber deception solution to improve network security. The system uses honeypots and decoy ports to divert attackers away from real services while safely observing their activities. By monitoring incoming network connections, the system can detect suspicious behaviors such as port scanning, repeated access attempts and unauthorized login attempts.

The project demonstrates how deception-based security can help organizations understand attacker behavior without exposing critical systems. DecoyShield records important attack details including IP addresses, accessed ports, timestamps and interaction patterns. This collected information provides valuable insights for threat analysis and future security improvements.

The adaptive mechanism of the system allows it to dynamically deploy additional honeypots when suspicious activity is detected, increasing the effectiveness of the deception strategy. The system also includes a simple monitoring dashboard that enables administrators to view alerts and attack information in real time.

Overall, DecoyShield provides a lightweight, cost-effective, and practical solution for enhancing network security. By combining cyber deception, continuous monitoring and attack analysis, the system helps protect real network resources while improving

awareness of potential cyber threats

7.2 Future Work

- **Integration of Machine Learning:** Future versions of DecoyShield can integrate machine learning algorithms to automatically detect advanced and unknown attack patterns. This will help the system analyze attacker behavior more intelligently and improve threat classification accuracy.
- **Scalability for Large Networks** Currently the system is suitable for small-scale or academic environments. In the future, it can be improved to support large enterprise networks with multiple distributed honeypots.
- **Cloud-Based Deployment:** DecoyShield can be enhanced by enabling cloud-based deployment, allowing organizations to monitor attacks across different locations and networks through a centralized platform.
- **Advanced Threat Intelligence Integration:** The system can be integrated with external threat intelligence databases to automatically identify malicious IP addresses, known attackers, and global cyber threats.
- **Distributed Honeypot Network:** A network of distributed honeypots across different systems or networks can be implemented to collect broader attack intelligence and improve cyber deception effectiveness.

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